Leo C. Stein

Contact Information

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Education

Ph.D., Physics, Massachusetts Institute of Technology, Cambridge, MA, USA **May 2012**

Dissertation Advisor: Prof. Scott Hughes

Dissertation Title: *Probes of strong-field gravity*

B.S., Physics, California Institute of Technology, Pasadena, CA, USA **June 2006**

Degree conferred with honor.

Senior Thesis Advisors: Dr. Patrick Sutton and Prof. Alan Weinstein

Employment

Associate Professor, University of Mississippi, Oxford, MS USA **July 2024–Present**

Assistant Professor, University of Mississippi, Oxford, MS USA **August 2018–June 2024**

Senior Postdoctoral Researcher, Caltech, Pasadena, CA USA **September 2015–August 2018**

NASA Einstein Fellow, Cornell, Ithaca NY, USA **September 2012–August 2015**

Research and Teaching Assistant, MIT, Cambridge MA, USA **September 2006–May 2012**

Teaching Assistant, Caltech, Pasadena, CA, USA **Fall 2004, Spring 2005**

Summer Research Fellow, Caltech, Pasadena, CA, USA **June–September 2003/2005**

Research Interests

General relativity (GR), gravitation, and astrophysical phenomena which can elucidate gravity. One major theme is pushing numerical and analytical gravitational-wave (GW) predictions to the precision frontier in advance of next-generation observatories. A second major theme is using GWs to test GR against beyond-GR models, in both theory-independent and theory-dependent models. This involves numerical relativity and renormalization methods applied to specific effective field models for beyond-GR theories.

Honors and Awards

Kavli Fellow, National Academy of Sciences, **2025**

Sloan Research Fellowship, Alfred P. Sloan Foundation, **2023–2025**

CAREER Award, NSF **2021–2026**

Einstein Postdoctoral Fellow, NASA **2012–2015**

Henry Kendall Teaching Award, Massachusetts Institute of Technology **2011**

Upperclass Merit Scholarship, California Institute of Technology **2005–2006**

Teaching Experience

Professor, University of Mississippi

Phys. 213, General physics I	**Spring 2021**
Phys. 3010, Intro to Phys Careers and Research	**Fall 2026**
Phys. 401, Electromagnetism I	**Falls 2019–2022**
Phys. 402, Electromagnetism II	**Springs 2019–2021**
Phys. 436, Intro to cosmology	**Falls 2023, 2025**
Phys. 463/4, Senior research project	**Fall 2020, Spring 2021, Fall 2023**
Phys. 503/630, Graduate reading course	**Spring 2019, Falls 2020–2021**
Phys. 709, Graduate classical dynamics I	**Fall 2018**
Phys. 721, Graduate electrodynamics I	**Springs 2022–2026**
Phys. 7360 (née 722), Graduate electrodynamics II	**Falls 2022–2026**
Phys. 735, General relativity	**Fall 2024**
Phys. 750, General relativity II	**Spring 2020**

Guest Lecturer, California Institute of Technology

Ph236, General relativity	**Fall 2017**
Ph237, Gravitational Waves	**Spring 2016**

Guest Lecturer, Massachusetts Institute of Technology

8.901, Graduate Astrophysics I	**Spring 2011**

Teaching Assistant, Massachusetts Institute of Technology

8.942, Cosmology	**Fall 2011**
8.901, Graduate Astrophysics I	**Spring 2011**
8.286, The Early Universe	**Fall 2009**

Teaching Assistant, California Institute of Technology

Ph 7, Nuclear and Quantum Physics Lab	**Spring 2005**
Ph 5, Analog Electronics for Physicists	**Fall 2004**

Mentoring/ Supervision

Postdoctoral researchers

Aniket Khairnar	**Fall 2026–Present**
Károly Csukás	**Fall 2021–Summer 2024**
Now a postdoc at the Wigner Research Centre for Physics	
José Tomás Gálvez Ghersi	**Fall 2019–Spring 2021**
Now faculty at Universidad de Ingeniería y Tecnología, Peru	

Graduate students

João Dinis Álvares, University of Mississippi	**Spring 2026–Present**
David Bronicki, University of Mississippi	**Fall 2019–Summer 2023**
Subhayu Bagchi, University of Mississippi	**Fall 2019–Summer 2026**
Aniket Khairnar, University of Mississippi	**Fall 2019–Summer 2026**
Akshay Khadse, University of Mississippi	**Fall 2018–Summer 2024**

Now a lecturer at Georgia College & State University

Lorena Magaña Zertuche, University of Mississippi **Fall 2018–Summer 2024**
Now a postdoc at NBI, Copenhagen, Denmark

Joe Rivest, University of Mississippi **Fall 2018–Summer 2024**

Sashwat Tanay, University of Mississippi **Fall 2018–Summer 2022**
Now faculty at Universidad de Ingeniería y Tecnología, Peru

Maria (Masha) Okounkova, Caltech **Fall 2015–Summer 2019**
Now faculty at Pasadena City College

Baoyi Chen, Caltech **Fall 2016–Summer 2018**

Undergraduate students

Wayne Zhao, Harvard **Summer 2016**
Princeton Ph.D.

Professional Activities, Outreach, and Service

LISA Consortium, Full member **2020–Present**
UMiss LISA Group leader **2020–Present**

Simulating eXtreme Spacetimes collaboration **2015–Present**
Executive committee member **2018–Present**

American Physical Society, lifetime member **2010–Present**
Division of Gravitational Physics

Secretary/Treasurer **2023–2026**

Executive Committee Member-at-Large **2016–2019**

Division of Astrophysics

Conference organizer

New frontiers in strong gravity, Benasque, Spain **July 2026**
Two week international conference, ∼60 participants

Gravitational Wave Physics and Astronomy Workshop 2025, Georgia Tech, **December 2025**
Week-long conference, ∼ 200 participants

Nonlinear Black Hole Perturbation Theory, U of Nottingham **September 2025**
Three day workshop, ∼ 70 participants

11th Gulf Coast Gravity Meeting (GCGM), UMiss **April 2025**
Two day conference, ∼ 50 participants

Simulating Extreme Spacetimes with SpEC and SpECTRE, ICERM **August 2024**
Week-long international workshop, ∼85 participants

New frontiers in strong gravity, Benasque, Spain **July 2024**
Two week international conference, ∼70 participants

Nonlinear Aspects of General Relativity, Princeton PCTS **October 2023**
Four day workshop, ∼100 participants

Numerical Relativity Community Summer School, ICERM **August 2022**
Week-long international summer school, 150 participants

New frontiers in strong gravity, Benasque, Spain **July 2022**
Two week international conference, 100 participants

Numerical Relativity beyond General Relativity, Benasque, Spain **June 2018**
Week-long international workshop, 59 participants

34th Pacific Coast Gravity Meeting (PCGM), Caltech **March 2018**
Two day conference, $\sim$ 125 participants

Unifying Tests of General Relativity, Caltech **July 2016**
Three day workshop, 52 participants

Seminar organizer

TAPIR seminar, Caltech **Fall 2015–Spring 2018**

General Relativity Informal Tea-Time Series (GRITTS), MIT **Fall 2011–Spring 2012**

MKI Journal Club, MIT **Fall 2007–Spring 2010**

Conference session chair; Judge for best student speaker award

April APS meeting, NY, NY **April 2022**

Midwest relativity meeting, Grand Rapids, MI **October 2019**

April APS meeting, Columbus, OH **April 2018**

34th Pacific Coast Gravity Meeting (PCGM), Caltech **March 2018**

33rd Pacific Coast Gravity Meeting (PCGM), UCSB **March 2017**

"April" APS meeting, Washington D.C. **January 2017**

32nd Pacific Coast Gravity Meeting (PCGM), CSU Fullerton **April 2016**

Theoretical Astrophysics in Southern California (TASC), CSU Fullerton **November 2015**

Journal referee

American Journal of Physics, Classical and Quantum Gravity, Journal of Cosmology and Astroparticle Physics, Journal of Open Source Software, General Relativity and Gravitation, Monthly Notices of the Royal Astronomical Society, Physics Letters B, Physical Review D, Physical Review Letters, Physical Review X, Reviews of Modern Physics, The Astrophysical Journal Letters, The Physics Teacher

Agency work

Reviewer for NSF, NASA

Outreach

CLA Lectures on Tap **October 2025**
Lecture: "Black Holes: The spooky truth about the ghosts of dead stars"

Oxford Science Café **April 2019**
Lecture: "The truth about black holes"

Guest on the *Starts With a Bang* podcast **March 25, 2019**
Episode 42: Black holes and gravitation

Invited speaker for Latin American Webinar on Physics **March 13, 2019**
Webinar 75: "Testing Einstein with numerical relativity"

Caltech astronomy public lecture series speaker **March 2018**
Lecture: "The truth about black holes"

Astronomy on Tap public lecture series speaker and volunteer **2016–2018**
Close to a monthly basis

Caltech astronomy public lecture series panelist and emcee **2016–2018**
Approximately every three months

Invited guest lecture on black holes and gravitational waves **November 2017**
Science of Space and Time, Hampshire College

Invited video Q&A session, public high school physics class **June 2017**
The Nova Project school, Seattle

Guest on *The Titanium Physicists Podcast*
Episode 80: Picturing the Bach Hole **August 21, 2019**
Episode 64: The edges of Einstein **April 25, 2016**
Episode 62: Black Bells **February 1, 2016**

Quora Q&A Session on gravitational waves and first detection **February 17, 2016**
83.9k+ views, 20.8k+ followers

Invited guest host, public screening of *COSMOS* with Q&A, **March/June 2014**
Science Cabaret/Cornell

Invited public talk at *Frontiers of Cornell Astronomy*, **November 2013**
Cornell Friends of Astronomy

Invited video chat, *Topics in Physics* course, **July 2013**
Stanford Education Program for Gifted Youth

Computer Skills

Expert in Mathematica, C/C++, Python, Bash. Proficient in Javascript. Experience in Haskell, Java, Julia. Expert at *nix and HPC. Markup languages: LaTeX, HTML, CSS, Markdown.

Software—Most contributions can be found at https://github.com/duetosymmetry. Member of the *Simulating eXtreme Spacetimes* (SXS) collaboration, contributor to the Spectral Einstein Code (SpEC). Member of the *Black Hole Perturbation Toolkit*. Author of `qnm` python package (https://github.com/duetosymmetry/qnm). Core collaborator on xAct (http://xact.es) abstract tensor calculus package for Mathematica. Coauthor of xTerior package for exterior differential geometry under xAct. Co-maintainer of community contributions at http://contrib.xact.es. Developed arXiv-keys browser extension/add-on for Chrome/Firefox. Author of `orcidlink` and coauthor of `gridpapers` packages for LaTeX.

Invited Talks

55. ASU physics department colloquium January 2026
54. Natl. U. of Singapore, "Math. Methods for the GR Two-body Problem" workshop August 2025
53. URI physics department colloquium November 2024
52. UNC physics department colloquium February 2024
51. UIUC astrophysics seminar December 2023
50. Harvard CMSA GR seminar October 2023
49. UMass Amherst, Amherst Center for Fundamental Interactions seminar September 2023
48. Albert Einstein Institute, "Connecting the Dots" panel discussion June 2023
47. Queen Mary Univ. of London, Gravitational memory workshop June 2023
46. Utah State University, Theoretical Physics Talks, March 2023
45. Iowa State, Physics and astronomy department colloquium, October 2022
44. UT Austin, Weinberg Institute seminar, October 2022
43. Vanderbilt, Physics and astronomy department colloquium, September 2022

42. ICERM, Advances in CS Classical and Quantum Gravity, May 2022
41. Flatiron CCA, Ringdown workshop, invited overview talk, February 2022
40. DAMTP (University of Cambridge), HEP/GR colloquium, January 2022
39. SISSA, Current challenges in gravitational physics workshop, April 2021
38. Flatiron CCA, Gravitational wave astronomy group seminar, January 2021
37. University of Birmingham, astrophysics seminar September 2020
36. Albert Einstein Institute, ACR division seminar July 2020
35. Black Hole Perturbation Toolkit, Spring 2020 workshop May 2020
34. American Physical Society Meeting April 2020
33. UVA, physics department colloquium November 2019
32. UT Dallas, physics department colloquium October 2019
31. Northwestern University, CIERA astrophysics seminar May 2019
30. ETH-ITS Zurich, "New horizons for gravity" workshop May 2018
29. UC San Diego, astrophysics seminar March 2018
28. UC Berkeley, 4D particle physics seminar March 2018
27. Kyoto University, YKIS2018a Symposium February 2018
26. Oakland University physics seminar February 2018
25. University of Wisconsin-Milwaukee gravity seminar January 2018
24. Caltech/JPL Gravitational-Wave (CaJAGWR) seminar January 2018
23. ICN UNAM, Relativity seminar December 2017
22. University of Mississippi, Astrophysics seminar November 2017
21. University of Florida, Astrophysics seminar November 2017
20. University of Nottingham, Mathematical Physics seminar July 2017
19. Sapienza University of Rome, New Frontiers in Gravitational-Wave Astrophysics June 2017
18. Rochester Institute of Technology, CCRG seminar March 2017
17. Penn State, IGC seminar March 2017
16. University of Mississippi, Strong Gravity/Binary Dynamics workshop February/March 2017
15. SUNY Stony Brook, "The universe through gravitational waves" December 2016
14. University of Pennsylvania, New Frontiers in Gravitational Radiation workshop December 2016
13. Cambridge MA, Event Horizon Telescope collaboration meeting November/December 2016
12. Northwestern University CIERA, "Fellows at the Frontiers" August/September 2016
11. Princeton University, GR@100++ panel discussion April 2016
10. Cambridge MA, Einstein fellows symposium October 2014
9. Perimeter Institute, Strong gravity seminar October 2014
8. Cornell University, Friends of astronomy outreach event November 2013
7. Cambridge MA, Einstein fellows symposium October 2013
6. SUNY Geneseo, Physics colloquium October 2013
5. University of Maryland, UMD gravity seminar October 2013
4. Yale University, YCAA seminar September 2013

3. Kyoto University, YITP long-term workshop June 2013
2. Cambridge MA, Einstein fellows symposium October 2012
1. Cornell University, Relativity lunch November 2011

Contributed Talks (selected)

25. 28th Capra Meeting on Radiation Reaction in General Relativity July 2025
24. 24th International meeting on GR (GR24) July 2025
23. American Physical Society Meeting April 2024
22. American Physical Society Meeting April 2023
21. LISA Symposium XIV July 2022
20. American Physical Society Meeting April 2021
19. American Physical Society Meeting April 2019
18. American Physical Society Meeting April 2018
17. Pacific Coast Gravity Meeting March 2017
16. American Physical Society Meeting ~~April~~ January 2017
15. Testing Gravity 2017 January 2017
14. 21st International meeting on GR (GR21) July 2016
13. American Physical Society Meeting April 2016
12. Eastern Gravity Meeting May 2015
11. American Physical Society Meeting April 2015
10. NEB 16 Recent developments in gravity September 2014
9. American Physical Society Meeting April 2014
8. XXVII Texas symposium on relativistic astrophysics December 2013
7. 20th International meeting on GR (GR20) July 2013
6. Eastern Gravity Meeting June 2013
5. American Physical Society Meeting April 2013
4. Caltech TAPIR Seminar December 2011
3. Eastern Gravity Meeting June 2011
2. American Physical Society Meeting April 2011
1. American Physical Society Meeting April 2010

References

Scott A. Hughes, Professor of Physics, Massachusetts Institute of Technology
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Éanna É. Flanagan, Professor of Physics and Astronomy, Cornell University
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Yanbei Chen, Professor of Physics, California Institute of Technology
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